ACT-R's Declarative Memory in CogLean

Laws, Two Explanations of Confident Errors, and the Power Law of Practice

Jeff Shrager

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Draft for Comment

CogLean, in one slide

- A library of cognitive theory in Lean, organized like Mathlib: **mechanisms** defined once; **phenomena** stated with no mechanism in them; **theorems** relating them, under explicit **conditions**; **laws** for whole classes of mechanisms.
- Theorems are **explanations**, proved about the same definitions that run.
- A value computed at one setting is a **witness**, not an explanation.

This talk: what follows from ACT-R's declarative memory.

1. Which **laws** it has, at every setting.
2. **Confident errors**: outside a class, yet approached; a second explanation.
3. The **power law of practice**: which one, under which schedule and decay.

ACT-R's declarative memory, formalized

- A chunk: a feature code and the times it was used. Retrieval: softmax over activations, or failure.
- In the rational regime (decay $d = 1$, temperature $s = 1$):

$$e^{A_i} = \left(\sum_k \frac{1}{t - t_k} \right) \mu^{m_i}, \qquad P(i) \propto e^{A_i}, \qquad P(\text{fail}) \propto \theta.$$

- **Latency**: Fe^{-A}. **Priming**: spreading activation multiplies a chunk's weight by e^S, a factor ρ.
- Failure always has weight: **no retrieval is ever certain**.

Left out: other decays and temperatures (except for learning curves, over the reals), retrieval as use, production rules.

Part 1. Which laws, at every setting

Law	ACT-R	what it means
Certainty	**never**	a just-studied item doesn't retrieve itself for certain
Stability	**never**	studying one item changes another's self-retrieval
Order-freedom	**never**	the order of study matters
Determinacy	**never**	retrieval has more than one outcome
Recency	**has**	the later of two is retrieved twice as often
Stored-only	**has**	only studied items are retrieved

For every positive failure weight and mismatch factor: verdicts, not samples.
Consequences: graded confusion; **interference** with practiced items (errors .29 $\rightarrow$ .58); an order effect (10/17, any guessing bias).

Part 2. Confident errors, and the class ACT-R isn't in

- **The phenomenon**, stated once: (i) the familiar q is never answered wrongly from memory; (ii) every answer to the stranger x from memory is wrong; (iii) x backs up exactly as often as q.
- Proved for **any memory** with determinacy, certainty and stability. EPAM's net has all three; ACT-R's has **none**.

Naming people	EPAM	ACT-R
stranger Carl called Bob	45/46	225/391
Carl called Ann	0	225/782
"don't know": Carl / Bob	1/46 / 1/46	107/782 / 229/2254

Not covered isn't the same as can't: ACT-R misnames too; (iii) fails.

A graded pattern, and priming

- **Graded**: if x calls up q, and q itself, each with probability $\geq 1 - \delta$, the pattern holds within 2δ. *No certainty needed.*
- **Priming** q by ρ shrinks everything else relative to q:
  for every $\delta > 0$ there is ρ_0 such that every $\rho \geq \rho_0$ gives the pattern within 2δ.
- Never certain, so the exact pattern is a **limit approached**, not a value reached.

Multi-digit: $150 + 100$ answered 200 by retrieval	
ACT-R, unprimed	$225/529 \approx .43$
ACT-R, $100 + 100$ primed $\times 10$	$2250/2599 \approx .87$
ACT-R, primed $\times 100$	$22500/23299 \approx .97$
EPAM	$45/46 \approx .98$

What primes: having just seen it

- A recent use raises a chunk's base level. Seeing y j times just before the test adds j to each of y's base-level sums.
- The mismatch factor depends only on the code, so that's **one factor for every probe**:

$$\rho \;=\; 1 + \frac{k\,j}{B}.$$

- So **recency is priming**, with the factor derived: enough recent sightings of q give the pattern within any δ.

Bob seen 10 times just before: Carl called Bob $275/299 \approx .92$; Carl's "don't know" $(1/26)$ near Bob's $(629/20654)$.

Two explanations, and what tells them apart

	EPAM	primed ACT-R
mechanism	discrimination	partial matching + priming
errors follow	**shared features**	**the context**
after seeing Bob 10 times: **Ann** called Bob	0	$2475/3082 \approx .80$

- Priming draws retrieval away from *every* other item, familiar ones too.
- **Graded laws** (tolerance 0 = the exact laws) give the pattern for a class. Primed ACT-R **can't** have graded certainty.
- One explanation is a **law of the memory**; the other is a fact about **the context**.

Part 3. The power law of practice, unbundled

improves $\supset$ diminishing returns $\supset$ diminishing *relative* returns

$\supset$ approximately a power law (within ε, over a range) $\supset$ exactly a power law

- Predicates on a curve, with exact rationals: a power law is **scale invariance**, $T(kn) = r\,T(n)$, $r = k^{-b}$.
- Power and exponential curves are **different** phenomena: no positive curve is both.
- **Averaging**: the mean of two exponential learners isn't exponential, and has a power curve's signature (Estes 1956; Heathcote et al. 2000).

Whether noisy data "look like" a power law is a statistical question; these are mathematical ones.

Decay 1: one mechanism, two schedules

- **Massed, tested after a delay** L: latency $= FL/n$. An **exact** power law, exponent 1.
- **Spaced, tested after each trial**: latency $= F/H_n$ (harmonic numbers).
  - improves, diminishing returns, diminishing relative returns: *every* qualitative signature;
  - but $H_{2n} \leq H_n + 1$ while $H_n \to \infty$: doubling ratio $\to 1$. **Not** a power law over long ranges; it's logarithmic.

n	1	2	4	8	16	32
$T(2n)/T(n)$	.67	.72	.77	.80	.83	.86

Yet over trials 1–8: within 7.5% of a power law with exponent $\approx .47$ (a witness).

Decay $d < 1$: Anderson's account, checked

- Needs real powers: a separate package, `ACTR/`, the only one using Mathlib.
- **Bridge**: CogLib's rational scale invariance is the real one; scale invariance at doubling is a power law with exponent $-\log_2 r$.
- **The sum lies between integrals**:

$$\frac{(1+n)^{1-d}-1}{1-d} \;\le\; \sum_{k\le n} k^{-d} \;\le\; 1+\frac{(1+n)^{1-d}-1}{1-d}.$$

- So the doubling ratio $\to 2^{-(1-d)}$: for every $\varepsilon > 0$, beyond some trial, spaced practice is within ε of a power law with **exponent** $1-d$.

Asymptotic: no explicit trial count yet.

What the practice results say

	massed, tested later	spaced, tested after each
decay $d = 1$	power law, exponent 1	logarithmic: **not** a power law
decay $d < 1$	power law, exponent 1	$\approx$ power law, exponent $1 - d$

- Whether there's a power law, and its exponent, depend on the **schedule** and the **decay**: conditions, not the mechanism alone.
- A curve can have every signature over a typical range without being one.
- The signature doesn't identify the mechanism: averaged exponentials have it too.

"Explains the power law of practice": say which member of the family, under which schedule and parameters.

Limitations, and next

Limitations

- Retrieval at decay 1, temperature 1; reals only for learning curves.
- No retrieval as use.
- Priming as a factor; recency derived, spreading activation not.
- Learning curves: exact-match latency only.
- The decay result is asymptotic.
- No fits to data: witnesses only.

Next

- ACT-R's model of the Siegler & Shrager addition data, as a rival account of the same data.
- Retrieval as use, through composed models.
- Spreading activation as a derived source of priming.
- Which families have the graded laws, at what tolerances.
- Other theories' learning curves; statistical predicates.

Take-home

- Stated once, ACT-R's memory yields **verdicts for every setting**: four laws never had, two always had.
- Confident errors: outside the class, yet **approached** by priming, and recency *is* priming. Two explanations, and an observation that separates them: **does priming make familiar items misnamed too?**
- One explanation is a law of the memory; the other a fact about the context.
- The power law of practice is several phenomena; ACT-R produces **different ones** under different schedules and decays.

Not formalizing proofs, but formalizing explanations,
so that we can reason about them.

`https://github.com/jeffshrager/CogLean`